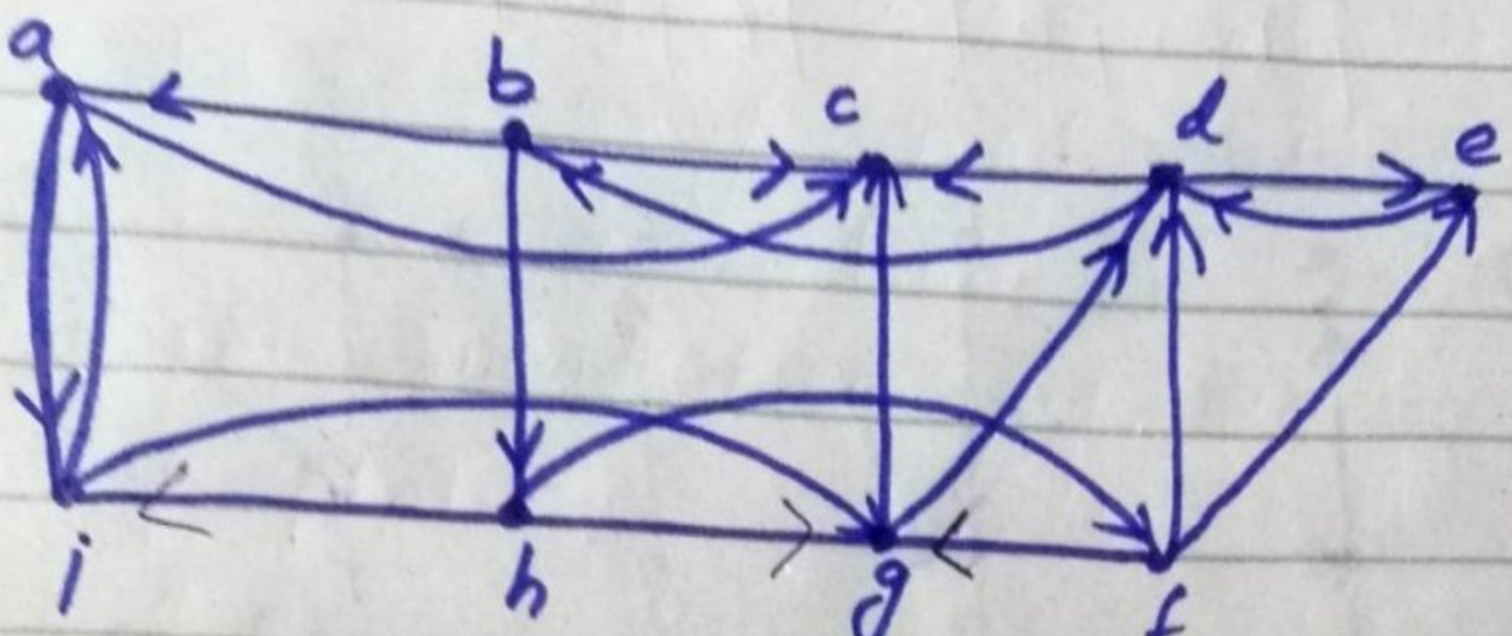
(c)



{a, b, c, d, e, f, g, h, i}, {c}

Exercise 9.2

2 List the ordered pairs in the relation R from $A = \{0,1,2,3,4\}$ to $B = \{0,1,2,3\}$ where $(a,b) \in R$ if and only if

(a) $a = b$

$\{(0,0), (1,1), (2,2), (3,3)\}$

(b) $a + b = 4$

$\{(3,3), (2,2), (3,1), (4,0)\}$

(c) $a > b$

$\{(1,0), (2,0), (2,1), (3,0), (3,1), (3,2), (4,0), (4,1), (4,2), (4,3)\}$

(d) $a \mid b$

Recall that $a \mid b$ means that b is a multiple of a (a is not allowed to be 0) $\frac{b}{a}$

$\{(1,0), (1,1), (1,2), (1,3), (2,0), (2,2), (3,0),$

Scanned by TapScanner

Scanned with CamScanner

4, 7, 7

(e) **gcd (a, b) = 1**

We need to look for pairs whose greatest common divisor is 1 in other words, pairs that are relatively prime. Thus the answer is ~~{(0,1), (1,0) (1,1), (1,2), (1,3), (2,0), (2,2), (3,0) (3,3), (4,0)}~~.

{(0,1), (1,0), (1,1), (1,2), (1,3), (2,1), (2, (3,1), (3,2), (4,1), (4,3)}

(f) **lcm (a, b) = 2**

There are not very many pairs of numbers (by definition only positive integers are concidered) whose least common multiple is 2 : only 1 and 2, and 2 and 2 Thus the answer is {(1,2), (2,1), (2,2)}

2:(a) List all the ordered pairs in the relation R = {(a,b) a divides b} on the set {1,2,3,4,5,6}

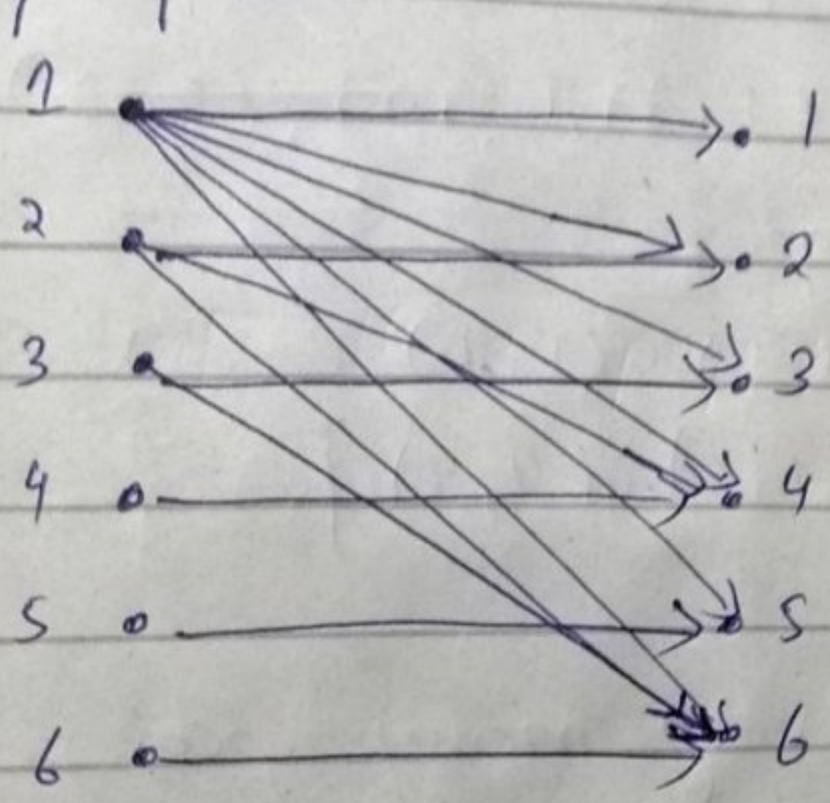
(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2) (2,4), (2,6), (3,3), (3,6), (4,4), (5,5) , (6,6)

(b) Display this relation graphically, as we dose in example 4.

Scanned by TapScanner

Scanned with CamScanner

We draw a line from a to b whenever a divides be, using seperate sets of points, an alternate form of this eg graph would have just one set of points.



(c) **Display this relation in tabulent form, as was done in e.g 4.**

We put an X in the ith and j^{th} column if and only if i divides j.

R	1	2	3	4	5	6
1	X	X	X	X	X	X
2		X		X		X
3			X			X
4				X		
5					X	
6						X

Scanned by TapScanner

Scanned with CamScanner

3 For each of these relations on the set {1, 2, 3, 4}, decide whether it is reflexive, whether it is symmetric whether it is antisymmetric and whether It is transitive

Reflexive

A relation R on a set A is called reflexive if $(a,a) \in R$ for every element $a \in A$

Note : Related to itself.

Transitive

A relation on a set A is called transitive if whenever $(a,b) \in R$ and $(b,c) \in R$ and $(a,c) \in R$ for all $a,b,c \in A$

Symmetric

if $(b,a) \in R$ then $(a,b) \in R$ for all $a,b \in A$

Antisymmetric

A relation R on a set A that for all $a,b \in A$ if $(a,b) \in R$ and $(b,a) \in R$ then $a = b$.}

(a) {(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)}

Transitive

(b) {(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)}

Reflexive, symmetric, Transitive

(c) {(2,4), (4,2)}

Symmetric

Scanned by TapScanner

Scanned with CamScanner

d $\{(1,2),(2,3),(3,4)\}$

Antisymmetric

e $\{(1,1),(2,2),(3,3),(4,4)\}$.

Reflexive, symmetric, antisymmetric, transitive.

(f) $\{(1,3),(1,4),(2,3),(2,4),(3,1)$
$(3,4)\}$

None of these properties.

4 Determine whether the relation R on the set of all people is reflexive, symmetric, antisymmetric and transitive where $(a,b) \in R$ if and only if

(a) a is taller than b

Being taller than is not reflexive (I am not taller than myself), not symetric (I am taller than my daughter but she is not taller than I). It is ~~only~~ antisymmetric (since we never have A taller than B, and B taller than A even if A=B). It is clearly transitive.

Scanned by TapScanner

Scanned with CamScanner

(b) a and b were born on the same day

This is clearly reflexive, symmetric and transitive (it is an equivalence relation — see section 9.5). it is not antisymmetric, since twins, for exemple, are unequal people born on the same day

(c) a has the same first name as b

This has exactly the same answer as part (b) since having the same first name is just like having same birthday

(d) a and b have a common grandparent

This is clearly reflexive and symmetric. It is not antisymmetric, since my cousin and I have a common grandparent and I and my cousin have a common grandparent but i am not equal to my cousin. This relation is not transitive. My cousin and I have a common grandparent my cousin and her cousin on the other side of her family have a common grandparent, my cousin's and I do not have common grandparent

Scanned by TapScanner

Scanned with CamScanner

5. Determine whether the relation R on set of all webpages is reflexive, symmetric, antisymmetric and transitive where $(a, b) \in R$ if and only if

(a) everyone who has visited web pages a has also visited web page b.

It is a tautology that every one who has visited web page a has also visited web page, so R is reflexive. It is not symmetric because there surely are web pages a and b such that set of people who visited a is a proper subset of the People who visited b.

Scanned by TapScanner

Scanned with CamScanner

6 Determine whether the relation R on the set of all real numbers is reflexive, symmetric, antisymmetric and transitive where $(x,y) \in R$ if and only if

(a) $x+y=0$

Since $1+1 \neq 0$, this relation is not reflexive since $x+y=y+x$ it follows that $x+y=0$ if and only if $xy+x=0$. so the relation is symmetric. since $(1,-1)$ and $(-1,1)$ are both in R, the relation is not antisymmetric. The relation is not transitive for e.g $(1,-1) \in R$ and $(-1,1) \in R$ but $(1,1) \notin R$.

(b) $x = \pm y$

$x = \pm x$ is reflexive. since $x = \pm y$ if and only if $y = \pm x$ the relation is symmetric. since $(1,-1)$ and $(-1,1)$ are both in R; the relation is not antisymmetric. The relation is transitive, essentially because the product of 1's and -1's is ± 1.

(c) $x-y$ is a rational number

It is reflexive since $x-x=0$ is a rational number.

Scanned by TapScanner

Scanned with CamScanner

It is symmetric because if $x-y$ is rational, then so is $(-x-y) = y-x$. Since $(1,-1)$ and $(-1,1)$ are both in R, the relation is not antisymmetric. To see that the relation is transitive, note that if $(x,y) \in R$ and $(y,z) \in R$ then $x-y$ and $y-z$ are rational number. Therefore their sum $x-z$ is rational, that means that $(x,z) \in R$.

$x = 2y$

$R = \{(x,y) \mid x = 2y\}$.

R is not reflexive; $x = 2x$ is only true at $x = 0$

R is not symmetric $2 = 2(1)$ $(x=2, y=1]$ while $1 = 2 \cdot 2$ is not true.

R is anti-symmetric if $x = 2y$ and $y = 2x$. then $x = y = 0$ since $x = 2y = 2(2x)$ $= 4x$ can only be true where $x = 0$

R is not transitive if $x = 2y$ and $y = 2z$ then $x = 2y = 2(2z) = 4z$ which is not equal to $x = 2z$.

Scanned by TapScanner

Scanned with CamScanner

(e) $xy \geq 0$

$R = \{(x,y) \mid xy \geq 0\}$

R is reflexive $xx = x^2 \geq 0$ is always true.

R is symmetric if $xy \geq 0$ then $yx \geq 0$

R is not antisymmetric as $(-2)(-1) \geq 0$

and $(-1)(-2) \geq 0$

while $-2 \neq -1$

R is not transitive if $x = -1, y = 0$ and

$z = 1$ these we get

$xy \geq 0 ; yz \geq 0 ; xz < 0$

~~(g) $xy = 0$~~ $x = 1$

Not reflexive since $(2,2) \notin R$

not symmetric since $(1,2) \in R$ but $(2,1) \notin R$

antisymmetric because if $(x,y) \in R$ and

$(y,x) \in R$ then $x = 1, y = 1$

so $x = y$.

Transitive because if $(x,y) \in R$ then

$(y,z) \in R$ and $x = 1$ & $y = 1$

so $(x,z) \in R$.

(f) $xy = 0$

Not reflexive since $(2,2) \notin R$

~~xxxx~~ symmetric Since (the roles of x and y in the statement are interchangeable

Scanned by TapScanner

Scanned with CamScanner

Not antisymmetric since (2,0) and (0,2) are both in R.

ZIve

NO Transitive for e.g (1,0) ∈R and (0,-2) ∈R but (1,-2) ∉R

h $x=y$ or $y=1$

Not reflexive since (2,2) ∉R.

symmetric the roles x and y is the statemen is interchangable

not symmetric since (2,1) and (1,2) are both in R

NOT Transitive for e.g (3,1) ∈R and (1,7) ∈R but (3,7) ∉R

QNO7: Determine whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric and transitive where (x,y) ∈R.

(a) $x \neq y$

symmetric

(b) $xy \geq 1$

symmetric, transitive.

(c) $x = y+1$ and $x = y-1$

symmetric.

(d) $x \equiv y \pmod 7$

Reflexive, symmetric, transitive

Scanned by TapScanner

Scanned with CamScanner

(e) x is multiple of y

Reflexive, transitive

(f) x and y are both negative or both nonnegative

Reflexive, symmetric, transitive

~~Reflexive, transitive~~

(~~f~~)

~~Reflexive, symmetric, transitive~~

(g) $x = y^2$

Antisymmetric

(h) $x \geq y^2$

Antisymmetric, transitive.

30 $R_1 = \{(1,2),(2,3),(3,4)\}$ and $R_2 = \{(1,1),(1,2),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3),(3,4)\}$ be relations from $\{1,2,3\}$ to $\{1,2,3,4\}$

Find

(a) $R_1 \cup R_2$

$\{(1,1),(1,2),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3),(3,4)\} = R_2$

(b) $R_1 \cap R_2$

$\{(1,2),(2,3),(3,4)\} = R_1$

(c) $R_1 - R_2$

$\emptyset$

Scanned by TapScanner

Scanned with CamScanner

(d) $R_2 - R_1$

$\{(1,1),(2,1),(2,2),(3,2),(3,3)\}$

(40) Let R_1 and R_2 be the divides and is a multiple of relations on the set of all positive integers, respectively That i's $R_1 = \{a,b\} / a$ divides b and $R_2 = \{(a,b) / a$ is a multiple of $b\}$ Find

(a) $R_1 \cup R_2$

The union of two relations is ~~intersection~~ union of these sets. Thus $R_1 \cup R_2$ ~~holds~~ b/w two integers if R_1 holds Or R_2 holds. Thus $(a,b) \in R_1 \cup R_2$. if and only if a/b or b/a There is not a good easier every to state this

(b) $R_1 \cap R_2$

The intersection of two relation is the intersection of these sets. Thus $R_1 \cap R_2$ holds between two if R_1 holds and R_2 holds. Thus $(a,b) \in R_1 \cap R_2$ if and only if a/b and b/a. This happens $a = \pm b$ and $a \neq 0$

Scanned by TapScanner

Scanned with CamScanner

(c) $R_1 - R_2$

By definition $R_1 - R_2 = R_1 \cap \bar{R}_2$. Thus this relation holds between two integers if R_1 holds and R_2 does not holds. Thus (a,b) we can write this in symbols by saying that $(a,b) \in R_1 - R_2$ if and only if $a|b$ and b equivalent to saying that $a|b$ and $a \neq \pm b$

(d) $R_2 - R_1$

By definition $R_2 - R_1 = R_2 \cap \bar{R}_1$. Thus this relation holds b/w two integers if R_2 holds and R_1 does not holds. We can write this in symbol by saying that $(a,b) \in R_2 - R_1$ if and only if $b|a$ and $a \nmid b$. This is equivalent to saying that $b|a$ and $a \neq \pm b$

(e) $R_1 \oplus R_2$

We know that $R_1 \oplus R_2 = (R_1 - R_2) \cup (R_2 - R_1)$ so we look at our solutions to part c and part d. Thus this relations holds b/w two integers if R_1 hold and R_2 does not hold or vice versa. This happens if and only if $a|b$ or $b|a$ but $a \neq \pm b$

Scanned by TapScanner

Scanned with CamScanner

Q41: Let R_1 and R_2 be the "congruent modulo 3" and "congruent modulo 4" relations respectively, on the set of integers. That is $R_1 = \{(a,b) \mid a \equiv b \pmod 3\}$ and $R_2 = \{(a,b) \mid a \equiv b \pmod 4\}$. Find

(a) $R_1 \cup R_2$

$\{(a,b) \mid a-b \equiv 0,3,4,6,8 \text{ or } 9 \pmod{12}\}$

(b) $R_1 \cap R_2$

Thus $(a,b) \in R_1 \cap R_2$ if and only if $a \equiv b \pmod 3$ and $a \equiv b \pmod 4$

Since $a-b$ is a multiple of both 3 and 4 if and only if $a-b$ is multiple of 12.

We can state that $a \equiv b \pmod{12}$

(c) $R_1 - R_2$

$(a,b) \in R_1 - R_2$ if and only if $a-b \equiv 3, 6 \text{ or } 9 \pmod{12}$

(d) $R_2 - R_1$

$(a,b) \in R_2 - R_1$ if and only if $a-b \equiv 4 \text{ or } 8 \pmod{12}$.

(e) $R_1 \oplus R_2$

$(a,b) \in R_1 \oplus R_2$ if and only if $a-b \equiv 3,4,6 \text{ or } 9 \pmod{12}$

Scanned by TapScanner

Scanned with CamScanner

Q42: List the 16 relation on the set {0,1}

These are just the 16 different relation of {(0,0),(0,1),(1,0),(1,1)}

1 $\emptyset$

2 {(0,0)}

3 {(0,1)}

4 {(1,0)}

5 {(1,1)}

6 {(0,0),(0,1)}

7 {(0,0),(1,0)}

8 {(0,0),(1,1)}

9 {(0,1),(1,0)}

10 {(0,1),(1,1)}

11 {(1,0),(1,1)}

12 {(0,0),(0,1),(1,0)}

13 {(0,0),(0,1),(1,1)}

14 {(0,0),(1,0),(1,1)}

15 {(0,1),(1,0),(1,1)}

16 {(0,0),(0,1),(1,0),(1,1)}

Q3 How many of the following 16 d/f relations on {0,1} contain the pair (0,1)?

A relation ~~on~~ a is just a subset. A subset can either contain a specified element or not, half of them do and half of them do not. Therefore

Scanned by TapScanner

Scanned with CamScanner

8 of 16 relations on {0,1} contain the pair (0,1). Alternatively a relation on {0,1} containing the pair (0,1) is just of the form $\{(0,1)\} \cup X$ where $X \subseteq \{(0,0),(1,0),(1,1)\}$ since this latter set has 3 element it has $2^3 = 8$.

Q44: Which of the following relations on {0,1}, which you listed in exercise 42

(a) Reflexive

8, 13, 14, 16.

(b) Irreflexive

1, 3, 4, 9

(c) Symmetric

1, 2, 5, 8, 9, 12, 15, 16

(d) antisymmetric

1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 13, 14

(e) asymmetric

1, 3, 4

(f) Transitive

1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15, 14, 16

Q45(a) How many relations are there on the set {a, b, c, d}

Scanned by TapScanner

Scanned with CamScanner

A relation on a set S with n elements is a subset of SxS. Since SxS has n^2 elements, we are asking for the number of subsets of a set with n^2 elements, which is 2^{n^2}. In our case $\boxed{n=4}$

So the answer is $\boxed{2^{4^2} = 2^{16} = 65,536}$

(b) How many relations are there on the set $\{a,b,c,d\}$ that contain the pair (a,a)?

In solving part (a), we had 16 binary choices to make - whether to include a pair (x,y) in the relation or not as x and y ranged over the set $\{a,b,c,d\}$. In this part, one of those choices has been made for us; we must include (a,a). We are free to make the other 15 choices.

So the answer is $\boxed{2^{15} = 32,768}$

47 How many relations are there on a set with n elements that are

(a) Symmetric

Scanned by TapScanner

Scanned with CamScanner

For each unordered pair $\{a,b\}$, whether include pairs $\{a,b\}$, $\{b,a\}$ or leave them out, this can be done in 2 ways for each unordered pair for each element $a \in A$, we need to decide whether to include (a,a) or not, again two possibilities we need to make 2-fold choice $C(n+1,2)$ times, since there are $C(n+2-1,2)$ ways to choose unordered pair with repetition allowed. Therefore answer is $2^{C(n+1,2)} = 2^{n(n+1)/2}$

(b) **Antisymmetric**

Similar to part (a)

For unordered pair $\{a,b\}$ of distinct element of A, we have 3 choice - either include (b,a) only or include neither

For each element of A we have 2 choice way

Therefore answer is $3^{C(n,2)} 2^n = 3^{n(n-1)/2} 2^n$

) **Asymmetric**

As part (b)

we have 3 way choice for $a \neq b$. There is no choice about including (a,a)

Scanned by TapScanner

Scanned with CamScanner

in the relation - the definition prohibits. The answer is $3^{C(n,2)} = 3^{n(n-1)/2}$

(d) **irreflexive**

For ordered pair (a,b) with $a \neq b$. There is no choice for pair (a,a) answer is $2^{P(n-2)} = 2^{n(n-1)}$

(e) **Reflexive and symmetric**

Just like part (a) except that there is no choice about including (a,a). For each unordered pair of distinct elements of A, we can choose to include neither or both of the corresponding ordered pair. Therefore the answer is $2^{C(n,2)} = 2^{n(n-1)/2}$

(f) **neither reflexive nor irreflexive.**

$$2^{P(n,2)}(2^n - 2) = 2^{n^2-n}(2^n - 2) = 2^{n^2} - 2^{n^2-n+1}$$

Q49: Find the error in the "proof" of the following "theorem"

Let R be a relation on a set A that is symmetric and transitive. Then R is reflexive

"Proof" Let $a \in A$. Take an element $b \in A$ such that $(a,b) \in R$. Because R is symmetric, we also have $(b,a) \in R$ Now using the transitive property,

Scanned by TapScanner

Scanned with CamScanner

We can conclude that $(a,a) \in R$ because $(a,b) \in R$ and $(b,a) \in R$.

The second sentence of the proof are "take an element $b \in A$ such that $(a,b) \in R$". There is no guarantee that such an element exist for the taking. This is only the mistake in the proof. if one could be guaranteed that each element in A is related to at least one element, then symmetry or transitivity would indeed inply reflexivity. Without this assumption, however the proof and the proposition is wrong

As simple example, take the relation $\emptyset$ on any nonempty set This relation is vacuously symmetric and transitive but not reflexive

Here another counter example the relation $\{(1,1),(1,2),(2,1),(2,2)\}$ on the set $\{1,2,3\}$

Scanned by TapScanner

Scanned with CamScanner

50 Suppose that R and S are reflexive relations on a set A
Prove or disprove each of these statements

(a) **R∪S is reflexive**
since R contain all the pairs (x,x) so does R∪S. Therefor R∪S is reflexive

(b) **R∩S is reflexive**
Since R and S each contain all the pairs (x,x), so does R∩S. Therefor R∩S is reflexive

(c) **R⊕S is irreflexive**
Since R and S each contain all pairs (x,x), we know that R⊕S contain none of these pairs. Therefore R⊕S is irreflexive

(d) **R−S is irreflexive**
Since R and S each contain all the pairs (x,x), R−S contain none of these pairs. Therefore R−S is irreflexive

(e) **S∘R is reflexive**
Since R and S each contain all the pairs (x,x), so does S∘R. Therefore S∘R is reflexive

56 Let R be the relation on the set $\{1,2,3,4,5\}$ containing the ordered pair
$(1,1), (1,2), (1,3), (2,3), (2,4), (3,1), (3,4), (3,5), (4,2), (4,5), (5,1), (5,2)$ and

Scanned by TapScanner

Scanned with CamScanner

Exercise 9.3

1: Represent each of those relations on $\{1,2,3\}$ with matrix.

a: $\{(1,1), (1,2), (1,3)\}$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

b: $\{(1,2), (2,1), (2,2), (3,3)\}$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

c: $\{(1,1), (1,2), (1,3), (2,2), (2,3) (3,3)\}$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

d: $\{(1,3), (3,1)\}$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Scanned with CamScanner

2: Represent each of these relations on $\{1,2,3,4\}$ with matrix.

a: $\{(1,2),(1,3),(1,4),(2,3),(2,4),(3,4)\}$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

b: $\{(1,1),(1,4),(2,2),(3,3),(4,1)\}$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

c: $\{(1,2),(1,3),(1,4),(2,1),(2,3),(2,4),(3,1),(3,2)$
$(3,4),(4,1),(4,2),(4,3)\}$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

d: $\{(2,4),(3,1),(3,2),(3,4)\}$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Scanned with CamScanner

3: List order pairs in relation on $\{1,2,3\}$ corresponding to these matrices.

a: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$\{(1,1),(1,3),(2,2),(3,1),(3,3)\}$

b: $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$\{(1,2),(2,2),(3,2)\}$

c: $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$\{(1,1),(1,2),(1,3),(2,1),(2,3),(3,1),(3,2),(3,3)\}$

4: List ordered pair in relation on $\{1,2,3,4\}$ corresponding to these matrices

a: $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

$\{(1,1),(1,2),(1,4),(2,1),(2,3),(3,2),(3,3),(3,4),$
$(4,1),(4,3),(4,4)\}$

Scanned with CamScanner

2 b: $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$

$\{(1,1),(1,2),(1,3),(2,2),(3,3),(4,1),(4,4)\}$

c: $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

$\{(1,2),(1,4),(2,1),(2,3),(3,2),(3,4),(4,1),(4,3)\}$

13: Let R be the relation represented by matrix.

$$M_R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Find the matrix reprecebing

a) R^{-1}

$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ $\because R^{-1} = R^t$

b) $\overline{R}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ $\because$ $0 \to 1$, $1 \to 0$

Scanned with CamScanner

c) R^2

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

14: Let R_1 and R_2 be relation on a set A represented by

$$M_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \text{ \& } M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

a) $R_1 \cup R_2$

matrix is formed by taking join $= M_{R_1 \vee R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

b) $R_1 \cap R_2$

matrix is formed by taking meet $= M_{R_1 \wedge R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

c) $R_2 \circ R_1$

matrix is Bolean Product $M_{R_2} \odot M_{R_1} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

d) $R_1 \circ R_2$

matrix is Bolean Product $M_{R_2} \odot M_{R_1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

e) $R_1 \oplus R_2$ exclusive-or

matrix is entrywise XOR $= \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ Same 0 different 1

Scanned with CamScanner

2

15: let R be relation represented by matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Find matrice that represent

a) R^2

$$R \cdot R = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

b) R^3

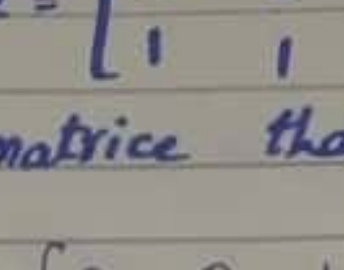
$$R^2 \cdot R = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

c) R^4

$$R^3 \cdot R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

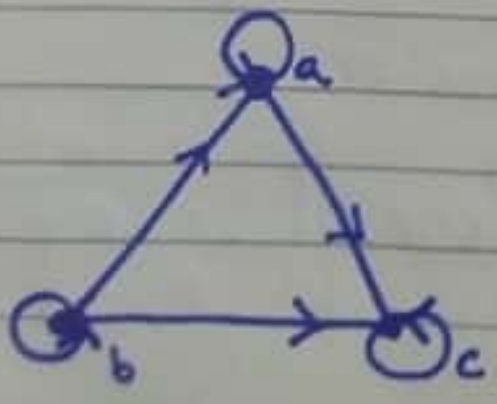
List the ordered pairs in relations represented by directed graphs.

23:



$\{(a,b),(a,c),(b,c),(c,b)\}$

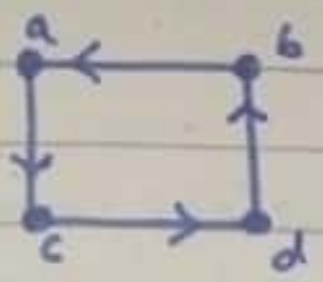
24:



Scanned with CamScanner

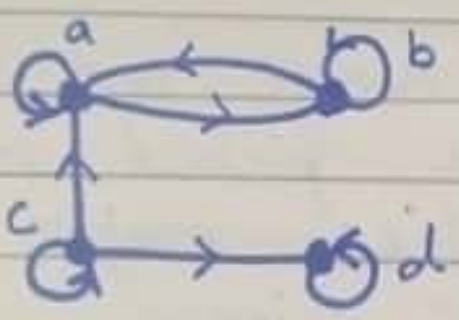
{(a,a),(a,c),(b,a),(b,b),(b,c),(c,c)}

25:



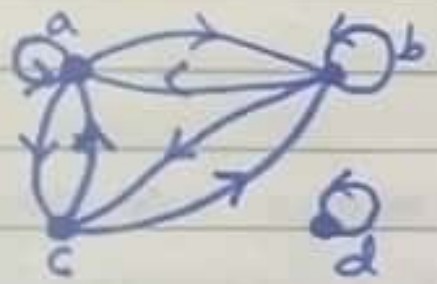
{(a,c),(c,d),(d,b),(b,a)}

26:



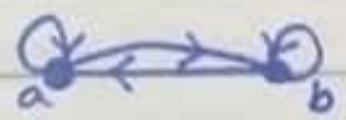
{(a,a),(a,b),(b,a),(b,b),(c,a),(c,c),(c,d),(d,d)}

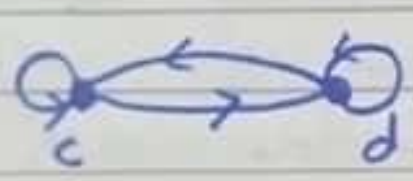
27:



{(a,a),(a,b),(a,c),(b,a),(b,b),(b,c),(c,a),
(c,b),(d,d)}

28





{(a,a),(a,b),(b,a),(b,b),(c,c),(c,d),(d,c),(d,d)}

Scanned with CamScanner

(5,4) Find

(a) R^2 (b) R^3 (c) R^4 (d) R^5

We just apply the definition each time

We find that R^2 contains all pairs in $\{1,2,3,4,5\} \times \{1,2,3,4,5\}$ except (2,3) and (4,5) and R^3, R^4 and R^5 contain all pairs

Exercise 9.4

1 Represent each of these relation on $\{1,2,3\}$ with a matrix

(a) $\{(1,1), (1,2), (1,3)\}$

1 Let R be the relation on the set $\{0,1,2,3\}$ containing the ordered pairs $(0,1), (1,1), (1,2), (2,0), (2,2)$ and $(3,0)$. Find

(a) Reflexive closure of R

The reflexive closure of R is R together with all pairs (a,a). Thus in this case the closure of R is

$\{(0,0), (0,1), (1,1), (1,2), (2,0), (2,2), (3,0), (3,3)\}$

Scanned by TapScanner

Scanned with CamScanner

Reflexive closure	symmetric closure	Transitive closure
• 9t is reflexive • 9t contain R • minimal relation satisfies (1) & (2)	• 9t is symmetric • contain R • minimal relation satisfies (1) & (2)	• 9t is transitive • 9t contain R • minimal relation satisfies (1) & (2)

(b) Symmetric closure of R.

The symmetric closure of R is R together with all pairs (b,a) for which (a,b) is in R. For example. since (1,2) is in R we need to add (2,1)

The closure of R is {(0,1),(0,2),(0,3) (1,0),(1,1),(1,2),(2,0),(2,1),(2,2) ,(3,0)}

3. Let R be the relation {(a,b) | a divides b} on the set of integers. What is the symmetric closure of R?

To form symmetric closure we need to add all the pairs (b,a) such that (a,b) is in R. In this case that means that we need to include pairs (b,a) such that a divides b, which is equivalent to saying that we need to include all the pairs (a,b) such that b divides a

Thus the closure is {(a,b) | a divides b or b divides a}

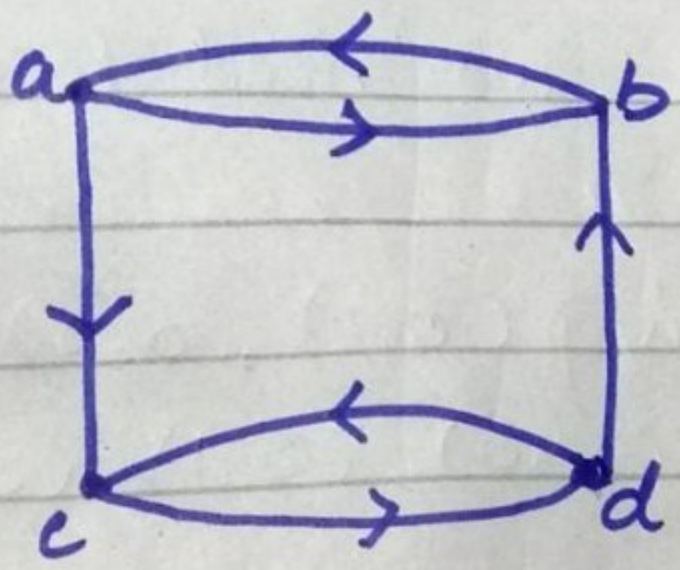
Scanned by TapScanner

Scanned with CamScanner

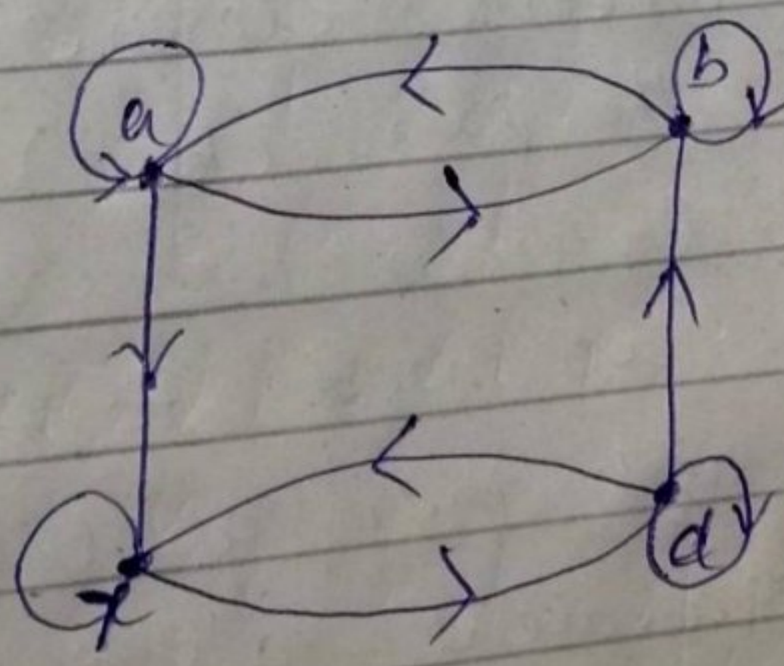
Q 5-7

Draw the directed graph of the reflexive closure of the relations with the directed graph shown

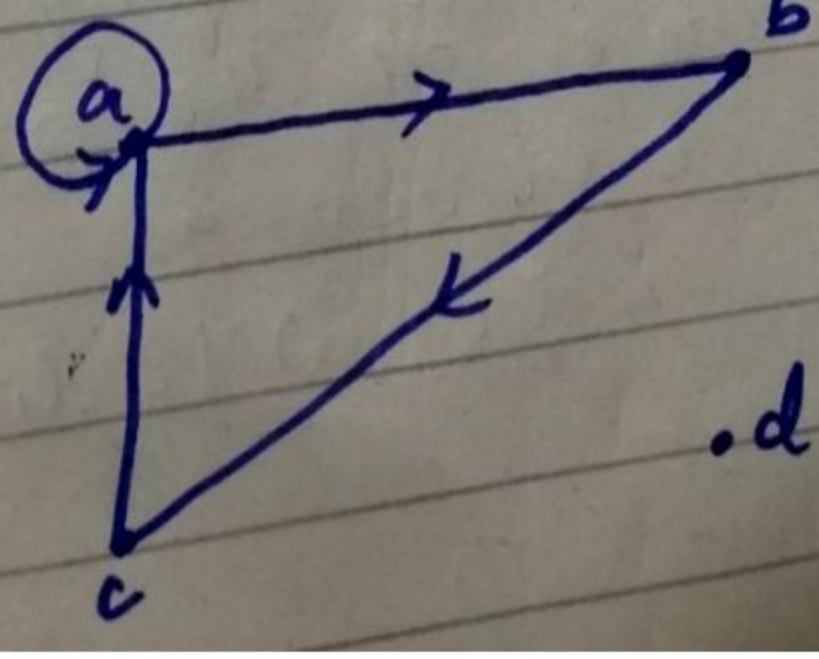
(5)



we form the reflexive closure by taking the given directed graph and appending loops at all the vertices at which there are not already loops

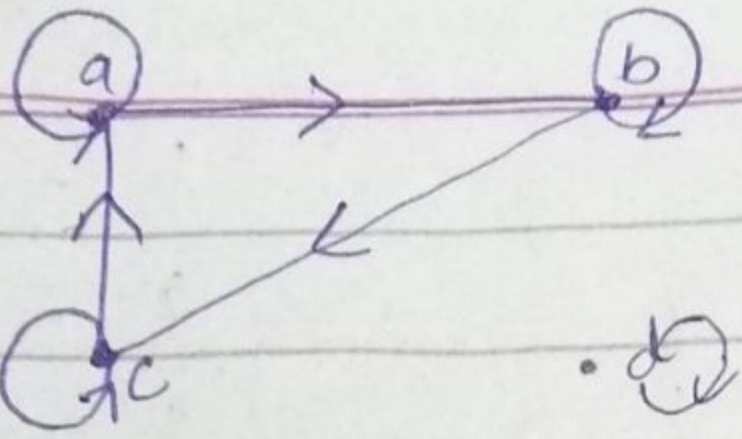


(6)

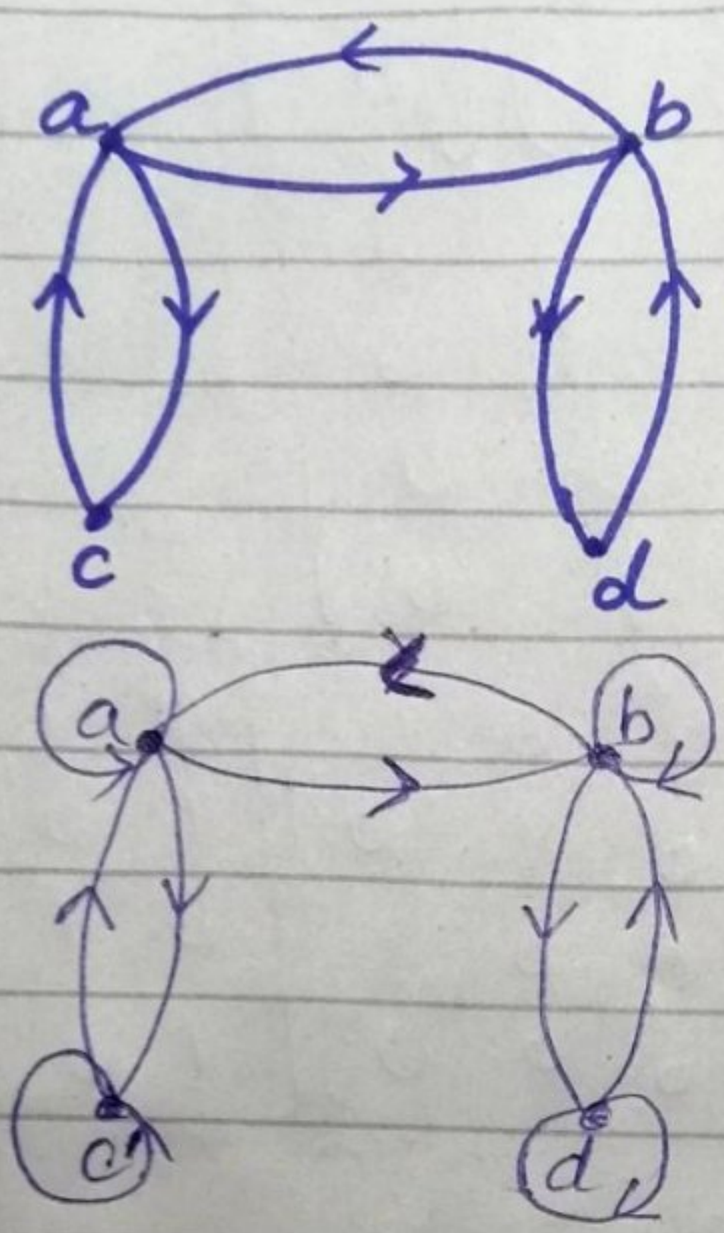


Scanned by TapScanner

Scanned with CamScanner



1



Q25: Use Algorithm 1 to find the transitive closures of these relations on {1,2,3,4}

(a) {(1,2),(2,1),(2,3),(3,4),(4,1)}

$$R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$R^{[2]} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Scanned by TapScanner

Scanned with CamScanner

$$R_3 = R_1 \cdot R_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$R_4 = R_1 \cdot R_3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$R_1 \vee R_2 \vee R_3 \vee R_4$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Scanned by TapScanner

Scanned with CamScanner

(b) $\{(2,1),(2,3),(3,1),(3,4),(4,1),(4,3)\}$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$R_1 \vee R_2 \vee R_3 \vee R_4$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

(c) $\{(1,2),(1,3),(1,4),(2,3),(2,4),(3,4)\}$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_1 \vee R_2 \vee R_3 \vee R_4$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Scanned by TapScanner

Scanned with CamScanner

$$= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(d) $\{(1,1),(1,4),(2,1),(2,3),(3,1),(3,2),(3,4),(4,2)\}$.

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$R_1 \vee R_2 \vee R_3 \vee R_4$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Q26: Use Algorithm 1 to find the transitive closures of these relations on $\{a, b, c, d, e\}$

(a) $\{(a,c),(b,d),(c,a),(d,b),(e,d)\}$

Scanned by TapScanner

Scanned with CamScanner

	a	b	c	d	e
a	0	0	1	0	0
b	0	0	0	1	0
c	1	0	0	0	0
d	0	1	0	0	0
e	0	0	0	1	0

$R_1 \vee R_2 \vee R_3 \vee R_4 \vee R_5$

$$\begin{bmatrix} 0&0&1&0&0 \\ 0&0&0&1&0 \\ 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&0&1&0 \end{bmatrix} \vee \begin{bmatrix} 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&1&0&0 \\ 0&0&0&1&0 \\ 0&1&0&0&0 \end{bmatrix} \vee \begin{bmatrix} 0&0&1&0&0 \\ 0&0&0&1&0 \\ 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&0&1&0 \end{bmatrix}$$

$$\vee \begin{bmatrix} 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&1&0&0 \\ 0&0&0&1&0 \\ 0&1&0&0&0 \end{bmatrix} \vee \begin{bmatrix} 0&0&1&0&0 \\ 0&0&0&1&0 \\ 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&0&1&0 \end{bmatrix}$$

$$= \begin{bmatrix} 1&0&1&0&0 \\ 0&1&0&1&0 \\ 1&0&1&0&0 \\ 0&1&0&1&0 \\ 0&1&0&1&0 \end{bmatrix}$$

$\{(b,c),(b,e),(c,e),(d,a),(e,b),(c,c)\}$

$$\begin{bmatrix} 0&0&0&0&0 \\ 0&0&1&0&1 \\ 0&0&0&0&0 \\ 1&0&0&0&0 \\ 0&1&1&0&0 \end{bmatrix}$$

Scanned by TapScanner

Scanned with CamScanner

$R \vee R_2 \vee R_3 \vee R_4 \vee R_5$

$$\begin{bmatrix} 0&0&0&0&0 \\ 0&0&1&0&1 \\ 0&0&0&0&1 \\ 1&0&0&0&0 \\ 0&1&1&0&0 \end{bmatrix} \vee \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&1&1&0&0 \\ 0&0&0&0&0 \\ 0&0&1&0&1 \end{bmatrix} \vee \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&0&1&0&1 \\ 0&0&0&0&0 \\ 0&1&1&0&1 \end{bmatrix}$$

$$\vee \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&1&1&0&1 \\ 0&0&0&0&0 \\ 0&1&1&0&1 \end{bmatrix} \vee \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&1&1&0&1 \\ 0&0&0&0&0 \\ 0&1&1&0&1 \end{bmatrix}$$

$$= \begin{bmatrix} 0&0&0&0&0 \\ 0&1&1&0&1 \\ 0&1&1&0&1 \\ 1&0&0&0&0 \\ 0&1&1&0&1 \end{bmatrix}$$

(c) $\{(a,b),(a,c),(a,e),(b,a),(b,c),(c,a),(c,b),(d,a),(e,d)\}$

$$\begin{bmatrix} 0&1&1&0&1 \\ 1&0&1&0&0 \\ 1&1&0&0&0 \\ 1&0&0&0&0 \\ 0&0&0&1&0 \end{bmatrix} \vee \begin{bmatrix} 1&1&1&0&1 \\ 1&1&1&0&1 \\ 1&1&1&0&1 \\ 0&1&1&0&0 \\ 1&0&0&0&0 \end{bmatrix} \vee \begin{bmatrix} 1&1&1&0&1 \\ 1&1&1&1&1 \\ 1&1&1&1&1 \\ 1&1&1&1&0 \\ 0&1&1&0&1 \end{bmatrix}$$

$$\vee \begin{bmatrix} 1&1&1&1&1 \\ 1&1&1&1&1 \\ 1&1&1&0&1 \\ 1&1&1&0&1 \\ 1&1&1&0&1 \end{bmatrix} \vee \begin{bmatrix} 1&1&1&1&1 \\ 1&1&1&1&1 \\ 1&1&1&1&1 \\ 1&1&1&1&1 \\ 1&1&1&0&1 \end{bmatrix}$$

Scanned by TapScanner

Scanned with CamScanner

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

(d) $\{(a,e),(b,a),(b,d),(c,d),(d,a),(d,c),(e,a),$
$(e,b),(e,c),(e,e)\}$.

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\vee \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Scanned by TapScanner

Scanned with CamScanner

EXERCISE 9.5

Equivalence Relation:-

A relation is equivalence if it is

1. Reflexive
2. Symmetric and
3. transitive

1. Which of these relations on {0,1,2,3} are equivalence relations? Determine the properties of an equivalence relation that the others lack?

a) {(0,0), (1,1), (2,2), (3,3)}

This is an equivalence relation; it is easily seen to have all the three properties.

The equivalence classes all have just one element.

b) {(0,0), (0,2), (2,0), (2,2), (2,3), (3,2)(3,3)}

This is not an equivalence relation.

because it is not reflexive (1,1) is missing also it is not transitive, since the pairs (0,2) and (2,3) are there but (0,3) is not there.

Scanned with CamScanner

Scanned with CamScanner

c) {(0,0), (1,1), (1,2), (2,1), (2,2), (3,3)}
This is an equivalence relation.

d) {(0,0), (1,1), (1,3), (2,2)(2,3)(3,1), (3,2), (3,3)}
This relation is reflexive and symmetric but not transitive, The pairs (1,3) and (3,2) are present but but not (1,2).

e) {(0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,2), (3,3)}
It is reflexive but not symmetric because (2,1) is missing so, it no E.R.

Q No 2:- Which of these relations on the set of all people are equivalence relations? Determine the properties of an equivalence Relation that the others lack?

a) {(a,b) | a and b are the same age}
This is an equivalence relation. bcs it's holds all the properties.

b) {(a,b) | a and b have the same parents}
This is an equivalence relation because it hold's all the three properties.

c) {(a,b) | a and b Share a common parent}
This is not an equivalence relation because it is not transive.
(If A is a child of W & X and B is a child of X & Y

Scanned with CamScanner

Scanned with CamScanner

and C is a child of Y and Z. So A and B are related to each other and B and C are also but A and C are not).

d) {(a,b) | a and b have met}

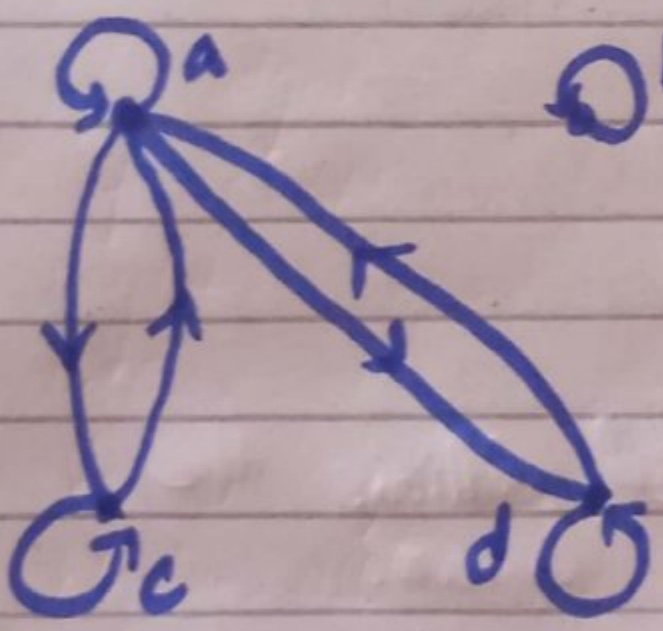
This is not an equivalence relation because it is not transitive.

e) {(a,b) | a and b speak a common language}

This is not an equivalence relation because it is not transitive.

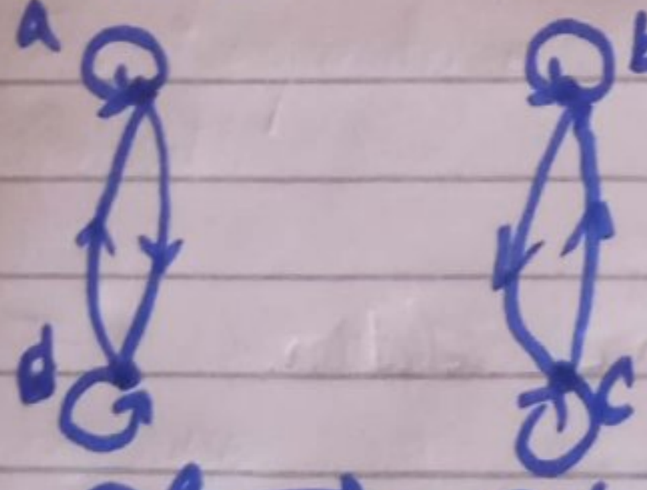
In Exercises 21-23 determine whether the relation with the directed graph shown is an equivalence relation.

21.



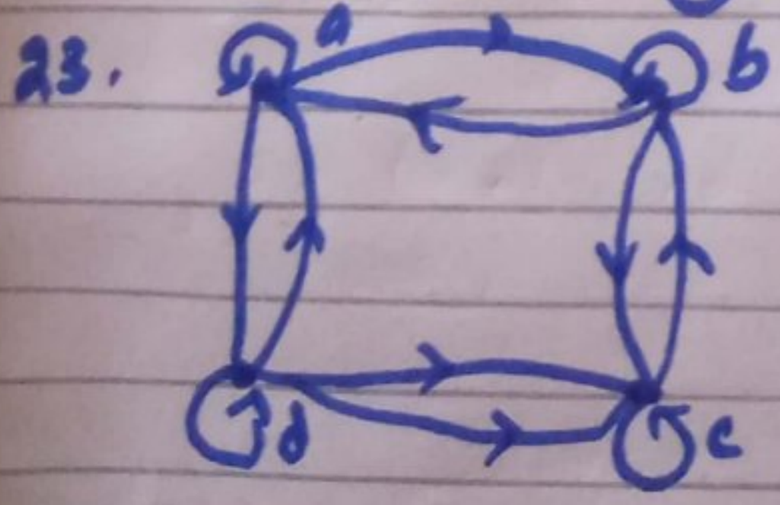
This is not an equivalence relation because it is not transitive, edges (c,d) and (d,c) is missing.

22.



This is an equivalence relation because it holds all the three properties.

23.



This is not equivalence relation because it is not transitive edge (a,c) is missing.

Scanned with CamScanner

Scanned with CamScanner

Q24:- Determine whether the relations represented by these Zero-one matrices are equivalence relations.

a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \neq A$

This is not an Equivalence relation since it is not Symmetric $A^T \neq A$.

b) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} = A$

This is an equivalence relation.

c) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

This is also an equivalence relation.

Scanned with CamScanner

Scanned with CamScanner

Exercise #9.6

1. Which of these relations on $\{0,1,2,3\}$ are partial orderings?

A relation is in partial ordering if it is reflexive, antisymmetric and transitive.

a) $\{(0,0),(1,1),(2,2),(3,3)\}$

given relation is in partial ordering because it holds all the properties.

b) $\{(0,0),(1,1),(2,0),(2,2),(2,3),(3,2),(3,3)\}$

This is not partial ordering because it not antisymmetric.

c) $\{(0,0),(1,1),(1,2),(2,2),(3,3)\}$

This is a partial ordering because $(1,2)$ is the only pair in the relation with the unequal components.

d) $\{(0,0),(1,1),(1,2),(1,3),(2,2)(2,2),(3,3)\}$

This is a "partial ordering because it is a "less than or equal to' relation on $\{0,1,2,3\}$. together with isolated point 0.

e) $\{(0,0),(0,1),(0,2),(1,0),(1,1),(1,2),(2,0),(2,2),(3,3)\}$

This is not a partial ordering bcs it is not antisymmetric.

2. Which of these relation on $\{0,1,2,3\}$ are PO?

a) $\{(0,0),(2,2),(3,3)\}$

It is not partial ordering bcs it is not reflexive $(1,1)$ is missing.

b) $\{(0,0),(1,1),(2,0),(2,2),(2,3),(3,3)\}$

This is partial ordering.

Scanned with CamScanner

Scanned with CamScanner

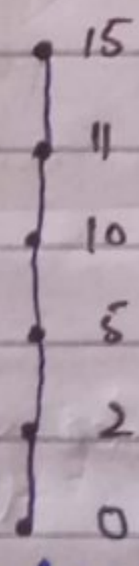
c) {(0,0), (1,1), (1,2), (2,2), (3,1), (3,3)}
This is not a partial ordering because it is not transitive. (3,1) and (1,2) is present but (3,2) is missing.

d) {(0,0), (1,1), (1,2), (1,3), (2,0), (2,2), (2,3), (3,0), (3,3)}
This is not a partial ordering because it is not transitive. (1,2) and (2,0) are present but (1,0) is missing.

e) {(0,0), (0,1), (0,2), (0,3), (1,0), (1,1), (1,2), (1,3), (2,0), (2,2), (3,3)}
This is not partial ordering because it is not antisymmetric, and not transitive.

21:- Draw the Hasse diagram for the "less than or equal to" relation on {0, 2, 5, 10, 11, 15}.

This is a totally ordered set, so the hasse diagram is linear.



22. Draw the hasse diagram for divisibility on the set.

Scanned with CamScanner

Scanned with CamScanner

a) {1,2,3,4,5,6}

Note that 1 divides all numbers, so the numbers on the second level from the bottom are primes.

A = {1,2,3,4,5,6}

R = "Divisibility"

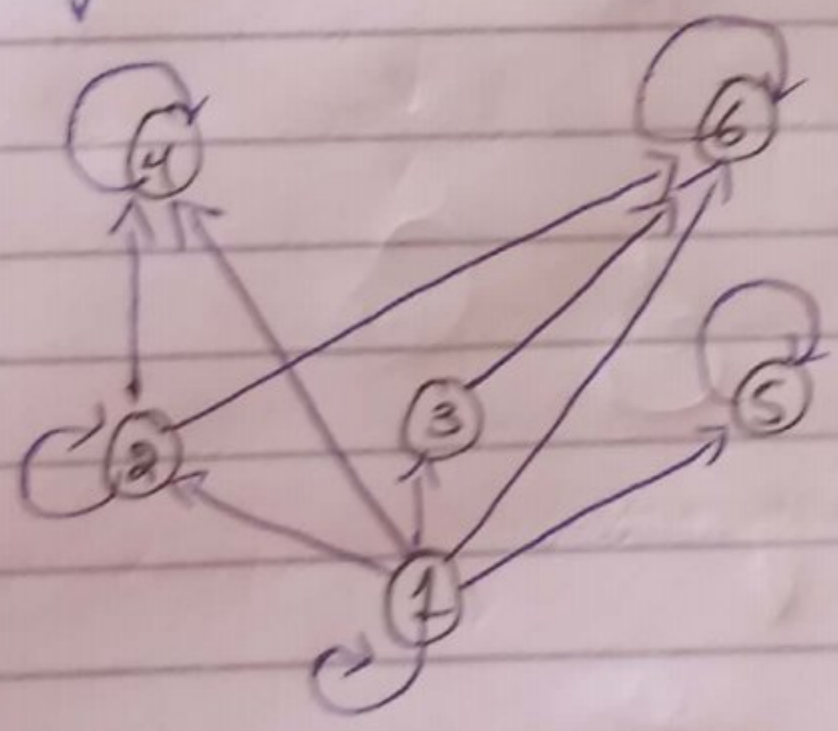
Steps to create Hasse diagram

Step 1: Create each vertex and then

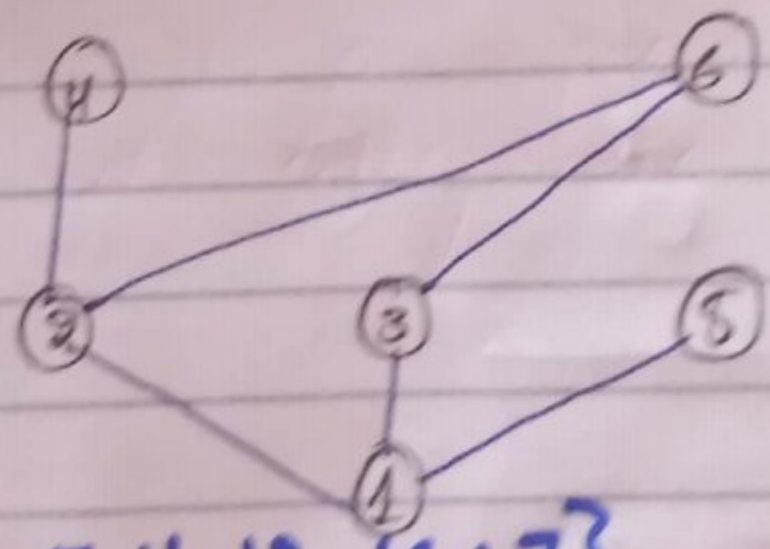
Step 2: Draw loops with itself of each vertex and anything else in the question.

Step 3: Remove self loops and transitive edges.

Solution:



Raw Hasse Diagram



Final Hasse diagram.

b) {3,5,7,11,13,16,17}

A = {3,5,7,11,13,16,17}

R = "Divisibility"

Scanned with CamScanner

Scanned with CamScanner

Note:- None of the integers in the given set are divisible by any other element in the list.

(3) (5) (7) (11) (13) (16) (7)

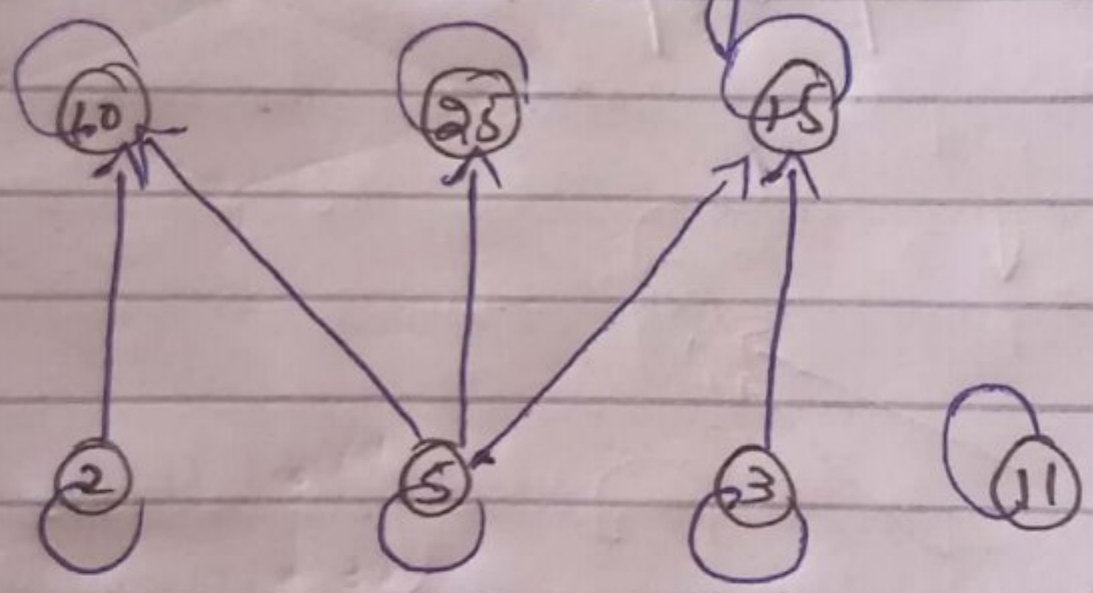
Find Hasse diagram:-

(3) (5) (7) (11) (13) (16) (7)

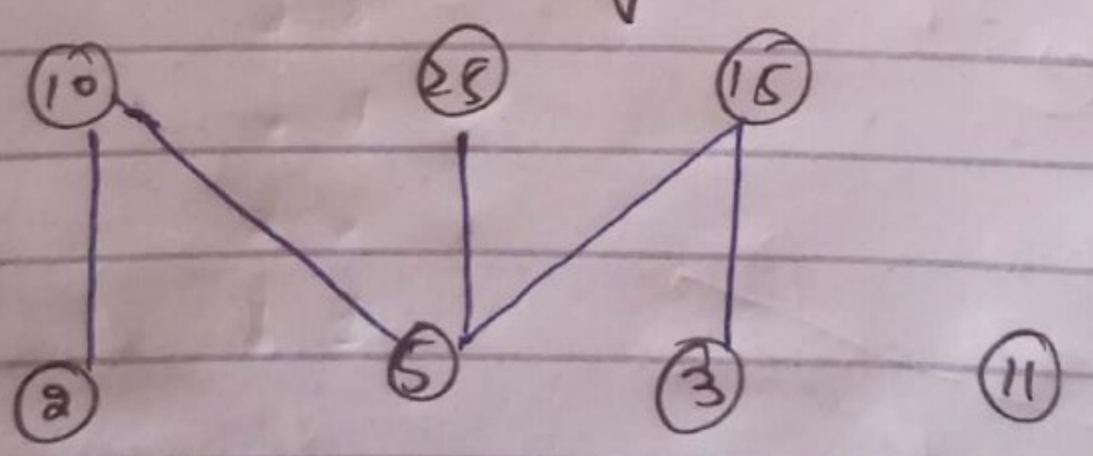
c) {2, 3, 5, 10, 11, 15, 25}

A_2 {2, 3, 5, 10, 11, 15, 25}

R_2 "Divisibility"



Final Hasse diagram:-



d) ~~{1, 2, 4, 8, 16, 32, 64}~~

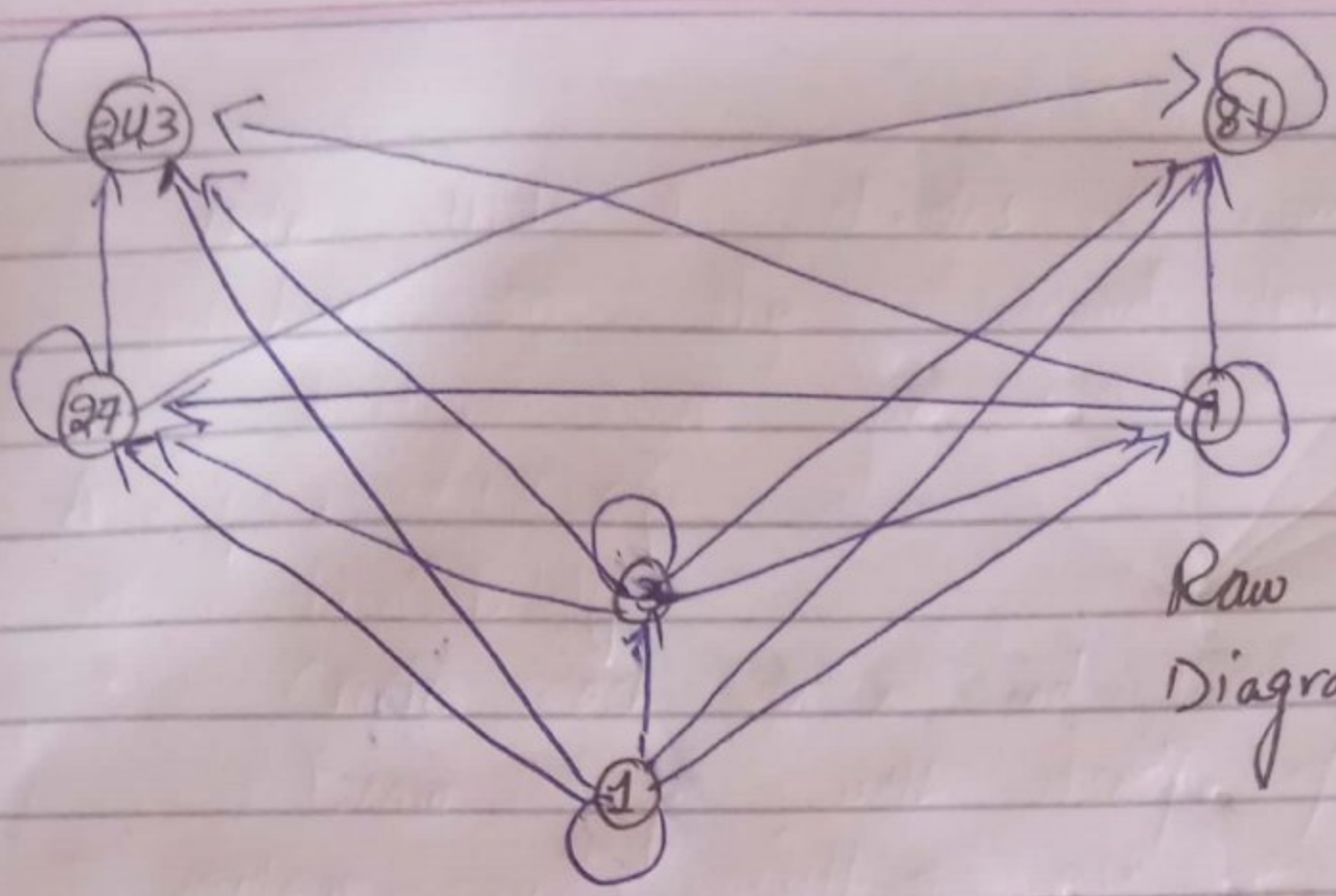
{1, 3, 9, 27, 81, 243}

A_2 {1, 3, 9, 27, 81, 243}

R_2 "Divisibility"

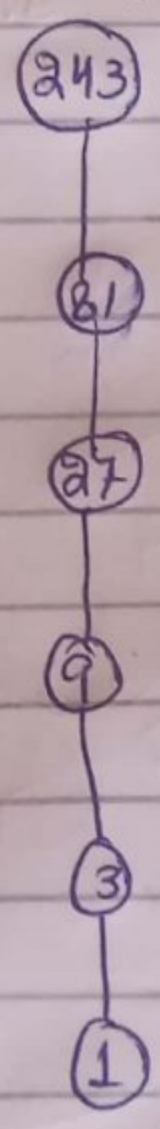
Scanned with CamScanner

Scanned with CamScanner



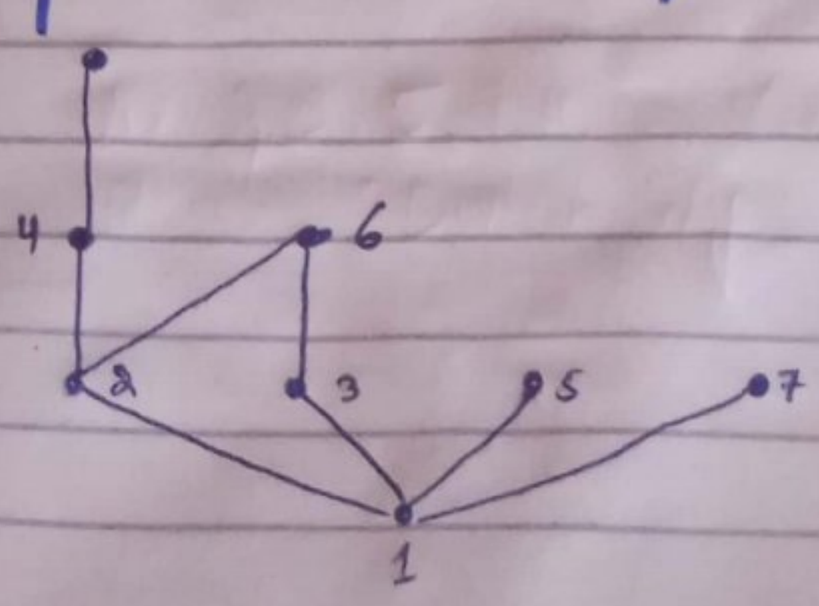
Raw Hasse Diagram.

Final Hasse diagram:-



23. Draw the Hasse diagram for divisibility on the set.

a) {1,2,3,4,5,6,7,8}



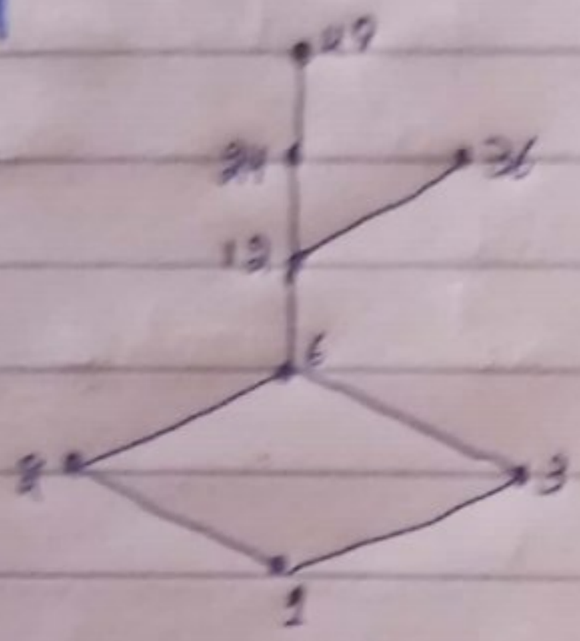
Scanned with CamScanner

Scanned with CamScanner

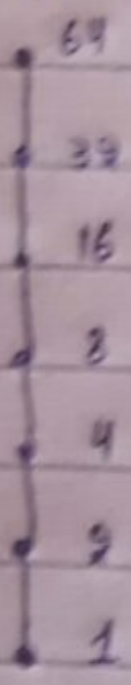
b) $\{1, 2, 3, 5, 7, 11, 13\}$



c) $\{1, 2, 3, 6, 12, 24, 36, 48\}$



d) $\{1, 2, 4, 8, 16, 32, 64\}$



Scanned with CamScanner

Scanned with CamScanner